

Course Objective(s)

2. To provide an understanding of the fundamental concepts of Artificial Intelligence
3. To provide the understanding of how machine learning can be applied to various real world applications.
4. To provide insight into applications of AI in various disciplines.

Course Outcomes

COs	Description
CO1	Describe the key concepts of AI
CO2	Explain machine learning concepts
CO3	Discuss about neural networks and its variants
CO4	Discuss about Text processing and language models
CO5	Discuss about applications of AI in various fields

CO-PO Mapping

PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO															
CO1	2	1	1	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	1	1	-	-	-	-	-	-	-	-	-	-	-	-
CO3	2	1	1	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	1	1	-	-	-	-	-	-	-	-	-	-	-	-
CO5	2	1	1	-	-	-	-	-	-	1	-	-	-	-	-

Syllabus**Unit I**

Introduction to AI- History and evolution, Key concepts and definitions, overview of Machine learning, neural networks, natural language processing

Unit II

Machine Learning Fundamentals: Supervised vs. unsupervised learning, Regression, classification, and clustering, Evaluation metrics and model validation

Unit III

Basics of neural networks: Introduction to deep learning: CNNs, RNNs, Applications in image and speech recognition

Unit IV

Natural Language Processing (NLP): Text preprocessing and representation, Language models and transformers, Applications in text generation, sentiment analysis, and machine translation

Unit V

AI in Healthcare and Biology: AI applications in medical imaging, diagnostics, and personalized medicine, Machine learning for genomics and drug discovery. AI in Social Sciences and Humanities: AI in economics: predictive analytics, market analysis, AI in psychology: behavioral modeling, cognitive simulations, Ethical and societal implications of AI

References

1. "Artificial Intelligence: A Modern Approach" by Stuart Russell and Peter Norvig.
2. Pattern Recognition and Machine Learning" by Christopher Bishop.
3. Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville.
4. Speech and Language Processing" by Daniel Jurafsky and James H. Martin.
5. The Ethics of Artificial Intelligence" by Nick Bostrom and Eliezer Yudkowsky

Evaluation Pattern

Assessment	Weightage (%)
Midterm	25
Continuous Assessment	25
End Semester Exam	50
Total Marks	100